

Assignment 5

SPARQL Queries

Consider the following RDF schema.

Classes

```
hs:Patient rdf:type rdfs:Class .  
hs:Encounter rdf:type rdfs:Class .  
hs:NurseNote rdf:type rdfs:Class .  
hs:ImageStudy rdf:type rdfs:Class .  
hs:Medication rdf:type rdfs:Class .
```

Properties

```
hs:hasEncounter rdf:type rdf:Property ;  
    rdfs:domain hs:Patient ;  
    rdfs:range hs:Encounter .  
hs:hasNote rdf:type rdf:Property ;  
    rdfs:domain hs:Encounter ;  
    rdfs:range hs:NurseNote .  
hs:hasImage rdf:type rdf:Property ;  
    rdfs:domain hs:Encounter ;  
    rdfs:range hs:ImageStudy .  
hs:prescribedMedication rdf:type rdf:Property ;  
    rdfs:domain hs:Encounter ;  
    rdfs:range hs:Medication .  
hs:patientName rdf:type rdf:Property ;  
    rdfs:domain hs:Patient ;  
    rdfs:range xsd:string .  
hs:noteText rdf:type rdf:Property ;  
    rdfs:domain hs:NurseNote ;  
    rdfs:range xsd:string .  
hs:noteDate rdf:type rdf:Property ;  
    rdfs:domain hs:NurseNote ;  
    rdfs:range xsd:date .  
hs:imageFinding rdf:type rdf:Property ;  
    rdfs:domain hs:ImageStudy ;  
    rdfs:range xsd:string .  
hs:medicationName rdf:type rdf:Property ;  
    rdfs:domain hs:Medication ;  
    rdfs:range xsd:string .
```

Write a SPARQL query for each of the following.

1

Find the names of all patients who have at least one nurse note.

```
SELECT DISTINCT ?patientName
WHERE {
  ?p a hs:Patient ;
    hs:patientName ?patientName ;
    hs:hasEncounter ?e .
  ?e hs:hasNote ?n .
}
```

2

Find all nurse notes written on 2026-03-20, and return the patient name and note text.

```
SELECT ?patientName ?noteText
WHERE {
  ?p hs:patientName ?patientName ;
    hs:hasEncounter ?e .
  ?e hs:hasNote ?n .
  ?n hs:noteDate "2026-03-20"^^xsd:date ;
    hs:noteText ?noteText .
}
```

3

Find all patients whose encounter has an image with finding “left opacity”.

```
SELECT ?patientName
WHERE {
  ?p hs:patientName ?patientName ;
    hs:hasEncounter ?e .
  ?e hs:hasImage ?i .
  ?i hs:imageFinding "left opacity" .
}
```

4

Write a natural-language question to express the following SPARQL query.

```
SELECT ?patientName ?noteText
WHERE {
  ?p a hs:Patient ;
```

```

        hs:patientName ?patientName ;
        hs:hasEncounter ?e .
    ?e hs:hasNote ?n .
    ?n hs:noteText ?noteText .
}

```

What are the names of the patients and the text of their associated nurse notes

5

Write a natural-language question to express the following SPARQL query.

```

SELECT ?patientName ?medName
WHERE {
    ?p a hs:Patient ;
        hs:patientName ?patientName ;
        hs:hasEncounter ?e .
    ?e hs:prescribedMedication ?m .
    ?m hs:medicationName ?medName .
}

```

Which medications have been prescribed to each patient, and what are the patients' names

Text Index

Consider five text nurse notes retrieved from the healthcare system.

Note ID	Nurse Note Text
N1	Patient reports chest pain and shortness of breath.
N2	Chest xray ordered after chest pain worsened overnight.
N3	Oxygen saturation dropped to 86 percent overnight.
N4	Patient denies chest pain today.
N5	Blood pressure improved after medication.

We take the following preprocessing:

1. all text is lowercased
2. punctuation is removed
3. tokenization is done by whitespace

For example, the processing converts N1 to the following output. N1 -> patient reports chest pain and shortness of breath

Tokenized N1: patient(1) reports(2) chest(3) pain(4) and(5) shortness(6) of(7) breath(8)

a

Construct and show the inverted index for the following terms:

- patient
- chest
- pain
- overnight
- oxygen

Word	Note ID
patient	N1, N4
chest	N1, N2, N4
pain	N1, N2, N4
overnight	N2, N3
oxygen	N3

b

Use the index to answer:

- Which notes match the Boolean query chest AND pain?
- Which notes match the Boolean query patient OR pain?
- Which notes contain patient OR pain but NOT chest?
- N1, N2, N4
- N1, N2, N4
- None

c

Construct and show the positional index for the terms:

- chest
- pain
- overnight

Word	Position
chest	N1[3], N2[1,5], N4[3]
pain	N1[4], N2[6], N4[4]
overnight	N2[7], N3[6]

Use the positional index to answer:

- Which notes contain the exact phrase chest pain?
- Which notes contain both pain and overnight?
- Which of those notes have pain appearing before overnight?
- N1, N2, N4
- N2
- N2

Code Practice: Image Retrieval and CLIP

You are given the HealthSense image retrieval notebook from class. Use it to explore how different retrieval methods behave. You may modify the notebook, run new queries, and change retrieval settings. For each part, include the code you changed, the output or screenshot, and a short explanation.

a

Compare retrieval methods. Choose one query image and one text query of your own. Play with the code, observe the results, and answer:

- Which answer looks most reasonable?
- Where is classical image retrieval weaker than CLIP?

CLIP performs better, it can more easily distinguish finer details whereas the regular methods can get confused with other things. Items changed were queried images in sections 5 and 10/11. For images 1-3 the regular method seemed to perform fine, but CLIP performed noticeably better on image 11.

b

Add at least two new synthetic HealthSense images. Then show the generated images, and test whether the retrieval system can find them appropriately using suitable queries.

Added:

```
("IMG013", "P112", "ED", "xray", "pleural effusion", "X-ray showing fluid at the base of the lungs."),
("IMG014", "P113", "Derm", "dermatology", "melanoma", "Dark, irregular lesion with jagged borders.")
```



to our sample

Queried "fluid in lungs"

```

if CLIP_READY:
    def clip_text_embeddings(texts):
        inputs = processor(text=texts, return_tensors="pt", padding=True, truncation=True).to(device)
        with torch.no_grad():
            feats = model.get_text_features(**inputs)
            feats = _ensure_tensor_output(feats)
            feats = feats / feats.norm(dim=-1, keepdim=True)
        return feats.cpu().numpy()

    def retrieve_by_text_clip(query_text, top_k=5):
        q = clip_text_embeddings([query_text])
        sims = cosine_similarity(q, X_clip)[0]
        order = np.argsort(-sims)[:top_k]
        out = df.iloc[order][["image_id", "modality", "finding", "unit", "report_text"]].copy()
        out["score"] = sims[order]
        return out

    display(retrieve_by_text_clip("fluid in lungs", top_k=5))
else:
    print("Skipping because CLIP is not ready.")

```

	image_id	modality	finding	unit	report_text	score
2	IMG003	xray	normal	ED	Chest X-ray with no focal consolidation.	0.288285
0	IMG001	xray	left opacity	ICU	Chest X-ray with left lung opacity and low oxygen concern.	0.285699
12	IMG013	xray	pleural effusion	ED	X-ray showing fluid at the base of the lungs.	0.280807
1	IMG002	xray	right opacity	ICU	Portable chest X-ray showing right-sided opacity.	0.279628
7	IMG008	microscopy	clustered cells	Lab	Microscopy field showing clustered blue cells.	0.274383

Our image was able to be retrieved.

C

Create three new natural-language queries for the CLIP text-to-image retrieval section, such as:

- a chest x-ray with opacity
- an ECG image with irregular rhythm
- a skin lesion with dark irregular borders

For each query, show the top retrieved results; explain why the results are reasonable; and explain one possible failure or ambiguity case.

Changed: text query in section 11

```

if CLIP_READY:
    text_query = "a chest x-ray with lung opacity"
    res_text = retrieve_by_text_clip(text_query, top_k=5)
    print("TEXT QUERY:", text_query)
    display(res_text)
    show_gallery(df[df["image_id"].isin(res_text["image_id"])], ncols=5)
else:
    print("Skipping because CLIP is not ready.")

```

... TEXT QUERY: a chest x-ray with lung opacity

	image_id	modality	finding	unit	report_text	score
2	IMG003	xray	normal	ED	Chest X-ray with no focal consolidation.	0.285548
0	IMG001	xray	left opacity	ICU	Chest X-ray with left lung opacity and low oxygen concern.	0.280391
1	IMG002	xray	right opacity	ICU	Portable chest X-ray showing right-sided opacity.	0.279431
12	IMG013	xray	pleural effusion	ED	X-ray showing fluid at the base of the lungs.	0.272316
5	IMG006	ecg	normal rhythm	Telemetry	ECG monitor with regular normal sinus rhythm.	0.238157

IMG001
xray | left opacity

IMG002
xray | right opacity

IMG003
xray | normal

IMG006
ecg | normal rhythm

IMG013
xray | pleural effusion

```

if CLIP_READY:
    text_query = "an ECG image with irregular rhythm"
    res_text = retrieve_by_text_clip(text_query, top_k=5)
    print("TEXT QUERY:", text_query)
    display(res_text)
    show_gallery(df[df["image_id"].isin(res_text["image_id"])], ncols=5)
else:
    print("Skipping because CLIP is not ready.")

```

... TEXT QUERY: an ECG image with irregular rhythm

	image_id	modality	finding	unit	report_text	score
4	IMG005	ecg	irregular rhythm	Cardiology	ECG strip with irregular rhythm; possible arrhythmia.	0.328264
5	IMG006	ecg	normal rhythm	Telemetry	ECG monitor with regular normal sinus rhythm.	0.317765
3	IMG004	ecg	tachycardia	Cardiology	ECG monitor image suggesting fast heart rate / tachycardia.	0.297622
1	IMG002	xray	right opacity	ICU	Portable chest X-ray showing right-sided opacity.	0.243187
2	IMG003	xray	normal	ED	Chest X-ray with no focal consolidation.	0.241260

IMG002
xray | right opacity

IMG003
xray | normal

IMG004
ecg | tachycardia

IMG005
ecg | irregular rhythm

IMG006
ecg | normal rhythm

```

if CLIP_READY:
    text_query = "a skin lesion with dark irregular borders"
    res_text = retrieve_by_text_clip(text_query, top_k=5)
    print("TEXT QUERY:", text_query)
    display(res_text)
    show_gallery(df[df["image_id"].isin(res_text["image_id"])], ncols=5)
else:
    print("Skipping because CLIP is not ready.")

```

... TEXT QUERY: a skin lesion with dark irregular borders

	image_id	modality	finding	unit	report_text	score
6	IMG007	microscopy	dense cells	Lab	Microscopy image with dense blue-stained cells.	0.245332
8	IMG009	microscopy	sparse cells	Lab	Microscopy field with sparse scattered cells.	0.245266
9	IMG010	dermatology	benign lesion	Derm	Dermatology photo of a round symmetric lesion.	0.238867
7	IMG008	microscopy	clustered cells	Lab	Microscopy field showing clustered blue cells.	0.234959
13	IMG014	dermatology	melanoma	Derm	Dark, irregular lesion with jagged borders.	0.229318

IMG007
microscopy | dense cells

IMG008
microscopy | clustered cells

IMG009
microscopy | sparse cells

IMG010
dermatology | benign lesion

IMG014
dermatology | melanoma

In most cases it was able to pull up some relevant images. Not all images were on topic, but for the most part was able to reasonably pick suitable images. The first two queries appeared to get EKGs confused with chest Xrays, this may seem reasonable as both are related to the chest. For the last query, the irregular borders may be confusing for the query as well, leading to the return of other non-skin samples.